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Extraction of DNA from canine feces for detection of *Campylobacter jejuni* using MagMax™ CORE Nucleic Acid Purification Kit on KingFisher™ DuoPrime Instrument

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We use this protocol and it's working

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Disclaimer

Reference to any commercial materials, equipment, or process does not in any way constitute approval, endorsement, or recommendation by the Food and Drug Administration.

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Abstract

Campylobacter jejuni is a significant cause of foodborne illness in humans and animals. Rapid detection of animals shedding *C. jejuni* is a key step in limiting the scope of outbreaks and reducing the potential exposure of humans and other animals. Manual DNA extraction methods, although effective, can be labor-intensive and time-consuming. The utilization of automated or semi-automated methods can significantly reduce the time needed for DNA extraction and significantly increase the number of samples that can be processed in a given amount of time. This method was developed to provide a standard method for semi-automated DNA extraction from feces for the detection of *C. jejuni* in veterinary patients.

Validation data (in-house and by an independent laboratory via collaborative study such as Randomized Blinded Method Test) are available upon request.



Guidelines

All manufacturers' intended use, safety information, storage guidelines, and recommendations should be followed.

Campylobacter jejuni is infectious to humans and is a biological safety level 2 organism. All work with specimens that potentially contain *C. jejuni* should follow the best practices outlined in the most current version of Biosafety in Microbiological and Biomedical Laboratories (BMBL) and institutional guidance.

Reporting of positive test results should follow all local, state, and federal regulations.

Follow the MagMaxTM CORE Nucleic Acid Purification Kit manufacturer's recommendations, warnings and procedural guidelines.

Materials

For DNA Extraction

1X PBS (Life Technologies cat#10010-023)

Applied Biosystems™ MagMAX™ CORE Nucleic Acid Purification Kit (ThermoFisher Scientific, cat#A32700/A32702)

KingFisher™ Duo Elution Strip, 40 pieces (ThermoFisher Scientific, cat# 97003520)

KingFisher™ Duo 12-tip comb, for Microtiter 96 Deepwell plate, 50 pieces (ThermoFisher Scientific, cat#97003500)

KingFisher™ Flex Microtiter Deepwell 96 plates (ThermoFisher Scientific, cat#95040460)

VetMAX™ Xeno™ Internal Positive Control DNA (ThermoFisher cat#s A29762/A29764)

VetMAX™ Xeno™ Internal Positive Control - VIC™ Assay (ThermoFisher Scientific A29765 or A29767)

From any manufacturer:

Microcentrifuge tubes - 1.5 mL, sterile

50 mL conical tubes - sterile

Pipette tips (1 mL, 200 µL, 20 µL and 10 µL)

Serological pipette, 25 mL

For PCR

TaqMan Fast Virus 1-step Master Mix (ThermoFisher Cat. Nos. 4444432/4444434/4444436)

Positive control DNA from *C. jejuni* ATCC 33291, ATCC 33292, or ATCC 33560

Negative control DNA from *E. coli* ATCC 25922

	A	B
	Oligo Name	5'-3' Sequence
	gyrA-F	AAGATACGGTCGATTTTGTTC
	gyrA-R	CTACAGCTATACCACTTGAACCATTTAATA
	gyrA-P	5'- [FAM]TGATGGTTCAGAAAGCGAACCTGATGTTTT[BHQ 1]-3'

Primer and probe sequences.

Primers and probe are diluted to a working concentration of 10 µM with a final concentration in the total reaction volume of 0.5 µM.

Equipment needed:

KingFisher™ Duo Prime Magnetic Particle Processor (ThermoFisher Scientific, cat#5400110)

Scale for weighing 8 g of feces

Pipettes for appropriate volumes

Centrifuge for microcentrifuge tubes capable of speeds of up to 16,000 X g

Vortexer for the fecal slurry



Vortexer with Qiagen (MoBio) Vortex Adapter (Cat#13000-V1-24) (optional)

Reference for primers and probe

Iijima Y, Asako NT, Aihara M, Hayashi K. (2004) Improvement in the detection rate of diarrhoeagenic bacteria in human stool specimens by a rapid real-time PCR assay. J Med Microbiol. 2004 Jul;53(Pt 7):617-22. PMID: 15184531

Protocol materials

⊗ VetMAX Xeno Internal Positive Control ND Thermo Fisher Scientific Catalog #A29762/A29764

⊗ 1X PBS Life Technologies Catalog #10010-023

⊗ MagMAX™ CORE Nucleic Acid Purification Kit Thermo Fisher Scientific Catalog #A32700/A32702

Safety warnings

! All manufacturers' intended use, safety information, storage guidelines, and recommendations should be followed.

Appropriate safety and personal protective equipment should be worn while working with chemicals including a suitable lab coat, disposable gloves, and protective eyewear.

Campylobacter jejuni is infectious to humans and is a biological safety level 2 organism. Work with specimens that may contain this organism should follow all applicable institutional guidelines.



Background

- 1 Fecal samples should be stored at 4-8 °C refrigerated until processed.
- 2 Samples should be **processed within 72 hours of receipt at the laboratory.**
- 3 Plates and enrichment broth should be incubated in a microaerophilic environment at 42 °C + or - 2 .
- 4 Remaining samples from the associated culture method can be used for molecular detection.
- 5 This protocol uses
 VetMAX Xeno Internal Positive Control ND Thermo Fisher Scientific Catalog #A29762/A29764
as a control for PCR inhibition.

Note

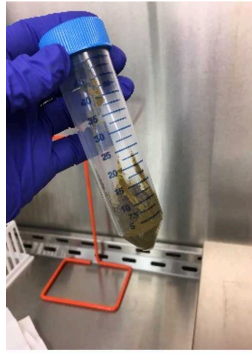
Note: This reagent can either be added to the CORE Lysis Solution at step 10 to serve as a DNA extraction control or after DNA extraction is complete to serve as the PCR inhibition control. If the laboratory has a system in place for DNA extraction and PCR inhibition, that method can be substituted in this protocol.

- 6 Prior to using this protocol, the script for the MagMAX™ CORE kit should be loaded onto the instrument.
The script is: **KingFisher DUO heated script: MagMAX_CORE_DUO.bdz**

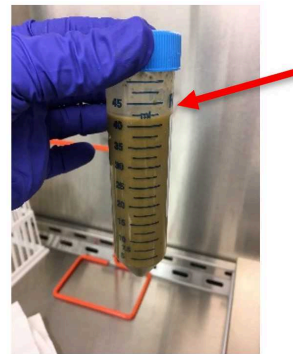
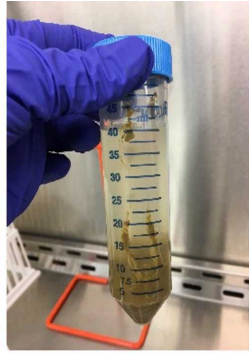
Sample preparation

2m 30s

- 7 Use 8 g of feces in 50 mL conical tubes
- 8 Add 32 mL 1X PBS Life Technologies Catalog #10010-023 to 8 g of feces. The feces in PBS will hereafter be called the feces-PBS slurry.



8 g feces



Add **32 mL PBS** to 8 g feces to prepare the feces-PBS slurry and allow the tube to sit for 2 minutes.

Vortex slurry for 30 seconds. Use slurry for direct PCR.

When transferring sample from the feces-PBS slurry pipette from the meniscus at the red arrow.

Making the Feces-PBS slurry.

8.1 Allow samples to sit for 00:02:00 after adding the 32 mL PBS

2m

8.2 Vortex samples for 00:00:30 to prepare fecal slurry

30s

PCR Method - DNA Extraction

1m

9 Extract DNA from the feces using the

MagMAX™ CORE Nucleic Acid Purification Kit **Thermo Fisher**
Scientific Catalog #A32700/A32702

following the manufacturer's protocol (outlined below).

10 Before processing the fecal slurry in Step 14, prepare the Bead-Proteinase K mix.

10.1 Vortex the MagMAX™ CORE Magnetic Beads thoroughly to ensure that the beads are fully resuspended.

10.2 Combine the following components for the required number of samples plus 10%.

A	B	C
Component	Volume per sample	Volume for 12 reactions + 10%
MagMAX™ CORE Magnetic Beads	20 µL	264 µL
MagMAX™ CORE Proteinase K	10 µL	132 µL
Total Bead/PK Mix	30 µL	

Table 1. Preparation of the Magnetic Beads and Proteinase K

Note

The manufacturer recommends that you prepare new Bead/PK mix for each processing run. If necessary, the Bead PK mix can be stored at  4 °C for up to 1 week.

11 Prepare the Lysis Solution.

11.1 Combine the following components for the required number of samples plus 10%.

A	B	C
Component	Volume per sample	Volume for 12 reactions + 10%
MagMAX™ CORE Lysis Solution	450 µL	5,940 µL
VetMAX™ Xeno™ Internal Positive Control DNA	4 µL	52.8 µL
Total Lysis Solution and Internal Positive Control DNA	454 µL	

Table 2. Preparation of the Lysis Solution

Note

Validation of the protocol used  4 μL VetMAX™ Xeno™ Internal Positive Control DNA per reaction instead of the  2 μL as noted in the manufacturer's protocol for consistency with prior manual extraction methods. Laboratories may choose to validate the use of  2 μL VetMAX™ Xeno™ Internal Positive Control DNA per reaction internally.

11.2 Mix by inverting the tube or bottle at least 10 times.

Note

(Optional) Store Lysis Solution at room temperature for up to 24 hours.

12 Prepare the Deep Well Sample and Wash Plate According to the indicated positions in the instructions and Table below

Add  30 μL of the bead mix to Row A of the Deep Well plate

Add  500 μL of MagMax Core Wash Solution 1 to Row B of the Deep Well plate

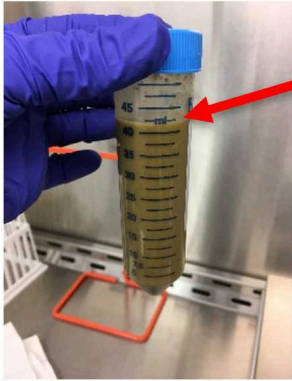
Add  500 μL of MagMax Core Wash Solution 2 to Row C of the Deep Well plate

In Row H, place a KingFisher™ Duo 12-tip comb, (ThermoFisher Scientific, cat#97003500)

	A	B	C	D
	Row ID	Row in the Plate	Reagent	Volume per well
	Sample	A	Bead mix	30 μL
	Wash 1	B	MagMAX™ CORE Wash Solution 1	500 μL
	Wash 2	C	MagMAX™ CORE Wash Solution 2	500 μL
	Tip Comb	H	Place a tip comb in the plate	

Table 3. Plate setup for the KingFisher™ Duo Prime instrument

- 13 Add $90\ \mu\text{L}$ the MagMAX™ CORE Elution Buffer to the Elution strip.
- 14 To a microcentrifuge tube, add $1\ \text{mL}$ of feces-PBS slurry from Sample Preparation step 7.



When transferring sample from the feces-PBS slurry pipette from the meniscus at the red arrow.

Pipetting the Feces-PBS slurry to a microcentrifuge tube.

- 15 Centrifuge the sample at $100 \times g$, Room temperature, 00:01:00

1m





16 Transfer  200 μL of the supernatant of the fecal slurry to a fresh 1.5 mL microcentrifuge tube.

17 Add  454 μL of the Lysis solution from step 11.

18 Vortex vigorously for  00:03:00 or until the sample is suspended.

3m

Note

Optional - The sample in the microcentrifuge tube can be placed in a vortexer with Qiagen (MoBio) Vortex Adapter (Cat#13000-V1-24)

19 Centrifuge the sample at  15000 x g, Room temperature, 00:02:00

2m



20 Without disturbing the pellet, transfer  200 μL of the the supernatant (clarified lysate) to row A of the Deep Well plate and mix by pipetting up and down several times.

21 Incubate the clarified lysate and bead mix for  00:02:00 at  Room temperature .

2m



- 22 Add  350 μL of MagMAX™ CORE Binding Solution to the clarified Isate-bead mixture in Row A.
- 23 Immediately proceed to processing the samples on the instrument by placing the Deep Well plate and the Elution strip in the KingFisher™ Duo Prime instrument
- 24 Select the appropriate script on the instrument. **KingFisher DUO heated script: MagMAX_CORE_DUO.bdz** 27m
 00:27:00
- Note

Processing on the machine takes approximately 27 minutes.
- 25 After samples have finished processing, place a cap on the elution strip and proceed with PCR or store samples at  4 °C until ready to perform PCR.

PCR Method - Setting up and running the PCR reactions

- 26 The probe is labeled with FAM with Black Hole Quencher 1 (BHQ1) so detection parameters appropriate for FAM or SYBR should be used. The developing laboratory uses Applied Biosystems™ VetMAX™ Xeno™ Internal Positive Control DNA (ThermoFisher Scientific A29762 or A29764) and the VetMAX™ Xeno™ Internal Positive Control - VIC™ Assay (ThermoFisher Scientific A29765 or A29767) to monitor for PCR inhibition.
- Note

Threshold is set at 0.2.
- 27 Create run on ABI 7500 Fast machine. Setting up the ABI 7500 Fast Thermocycler: Refer to the manual provided for running the machine. Reaction times are as follows in Table 1.

Equipment

7500 Fast Real-Time PCR System, desktop

NAME

Applied Biosystems

BRAND

4351107

SKU

	A	B	C	D	E	F
	Stage	Function	Temperature	Duration (sec)	Repeats	Optics
		Reverse Transcription	50°C	300		
	Stage 1	Initial PCR Activation	95.0°C	20		Off
	Stage 2	Annealing	95.0°C	3	Repeats 45 times	Off
	Stage 2	Extension	60°C	60		On

Table 4a. PCR amplification conditions in ABI 7500 under standard conditions are as follows:

	A	B	C	D	E	F
	Stage	Function	Temperature	Duration (sec)	Repeats	Optics
		Reverse Transcription	50°C	300		
	Stage 1	Initial PCR Activation	95.0°C	20		Off
	Stage 2	Annealing	95.0°C	3	Repeats 45 times	Off
	Stage 2	Extension	60°C	60		On

Table 4b. PCR amplification conditions in ABI 7500 under FAST conditions are as follows:

Note**Passive reference dye should be set as ROX**

- 28 Calculate the number of PCR reactions to be performed. Using Table 2 as a guide, determine the quantity of each reagent needed. Include a positive (*C. jejuni* ATCC 33291, ATCC 33292, or ATCC 33560) and negative (*E. coli* ATCC 25922) control and a "No Template" negative control.

For calculations, reference **Table 5**. PCR Worksheet for Reagent Volumes for Master Mix Preparation for *Campylobacter jejuni* PCR.

 Table 5. PCR Worksheet for Reage...

- 29 Combine reagents except for the DNA template in a 1.5ml or 2ml microcentrifuge tube and mix well.
- 30 Aliquot  18 μ L into each PCR tube or reaction well if using a plate.
- 31 Add samples and positive (*C. jejuni* ATCC 33291, ATCC 33292, or ATCC 33560 DNA) and negative (*E. coli* ATCC 25922 DNA) controls along with PCR inhibition controls
-  VetMAX Xeno Internal Positive Control ND **Thermo Fisher Scientific** Catalog #A29762/A29764
- .
- 32 Place Samples in thermocycler and perform run.
- 33 Following run, analyze the results. Results should be reported with a Ct value and interpretation.
- 34 Samples with a Ct value of 40 or less should be interpreted as positive. Samples with a Ct value >40 or an undetermined Ct value should be considered negative.



	A	B	C
	Target	CT Value	Interpretation
	gyrA	<=40	Detected
		>40	Not detected

Table 6. Interpretation of Ct Value

A	B
Oligo Name	5'-3' Sequence
gyrA-F	AAGATACGGTCGATTTTGTTC
gyrA-R	CTACAGCTATACCACTTGAACCATTTAATA
gyrA-P	5'-[FAM]TGATGGTTCAGAAAGCGAACCTGATGTTTT[BHQ1] 3'

Table 7. Primers and Probe